Abhiraj Singh

Lead AI Engineer · Agentic AI & Automation · LLM Fine-Tuning · RAG · AI Enablement

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PROFESSIONAL SUMMARY

Lead AI Engineer building agentic systems and AI automation for aerospace and defense manufacturing, from fine-tuned LLMs running in air-gapped environments to end-to-end SDLC and business-workflow automation. Build AI agents, automation tools and Claude skills for technical and non-technical teams alike, and teach every employee, technical or not, to use AI effectively.

PROFESSIONAL EXPERIENCE

iBase-t — Ahmedabad, India · On-site

Maker of Solumina MES, the manufacturing execution system that runs production for aerospace and defense manufacturers.

Lead AI Engineer — Sep 2026 – Present

- Lead AI engineering at iBase-t, setting direction for agentic systems, AI automation and LLM deployment across air-gapped and standard environments; building and mentoring a team of AI engineers.
- Design and ship workflow automations, automation tools, AI agents and Claude skills that remove repetitive manual work for technical and non-technical teams across the company.
- Run company-wide AI enablement, teaching technical and non-technical employees hands-on to get the most out of AI tools, agents and automation.
- Partner with product and engineering to embed AI into Solumina products and define on-premise LLM reference architectures.

AI Engineer — Jul 2025 – Sep 2026

- As the India office's sole AI engineer, built an AI assistant that carries out Solumina's manual workflows through natural language, powered by an in-house fine-tuned open-source LLM and a custom LangGraph agentic architecture with tool calling.
- Delivered production-grade LLM behavior in fully air-gapped environments: fine-tuned (LoRA/QLoRA) and served open-source models on-premise with zero cloud-API dependency, tuning for accuracy, latency and constrained hardware.
- Researched and prototyped multiple RAG architectures, including HippoRAG and knowledge-graph-based RAG, over technical manufacturing documentation; built retrieval on pgvector and benchmarked Pinecone, Qdrant, Chroma and FAISS.
- Built a complete AI-driven SDLC automation for internal engineering teams, covering requirements, code generation, review, testing and documentation, with evaluation and guardrails (golden test sets, grounding checks) to keep outputs reliable.

Sabudh Foundation — Mohali, India · Hybrid

Data Scientist Intern — Jan 2025 – Jun 2025

- Built a RAG document Q&A system on locally hosted LLMs (Ollama) with voice responses, and a Perplexity-style AI search interface combining retrieval with fine-tuned models.
- Fine-tuned domain-specific LLMs for accuracy and inference speed, and prototyped multi-modal chatbots.

Care.fi — Gurugram, India · On-site

Software Development Engineer Intern — Apr 2024 – Oct 2024

- Contributed to a healthcare insurance claims platform now used by hospitals across India, cutting a historically multi-day claims process to a few hours per patient.
- Owned the full stack of a new product (frontend, backend services and infrastructure), including a high-performance caching system built from the ground up; also built a PDF filter and editor for document workflows.

Deep Eigen — Remote

Frontend Developer Intern — Apr 2023 – Sep 2023

- Built a Django-based appointment system for new-recruit scheduling and revamped the front end of the Swaayatt website.

GeeksforGeeks Student Chapter (GFG CUIET), Chitkara University — Rajpura, India

Tech Team Coordinator — Jun 2022 – Mar 2024

- Led development of the chapter website (React, Node.js) and coordinated student tech teams.

TECHNICAL SKILLS

Generative AI	Agentic AI, Multi-Agent Systems, RAG (GraphRAG, HippoRAG), LLM Fine-Tuning (LoRA, QLoRA, PEFT), Tool Calling, Model Context Protocol (MCP), Prompt Engineering, LLM Evals & Guardrails
AI Automation	SDLC & Business Workflow Automation, AI Agents & Automation Tools, Claude Skills, Claude Code, Microsoft Copilot Studio
AI Frameworks	LangGraph, LangChain, Hugging Face, PyTorch, vLLM, Ollama, OpenAI, Anthropic & Gemini APIs
Vector & Data	pgvector, PostgreSQL, Pinecone, Qdrant, Chroma, FAISS, Knowledge Graphs, MongoDB, Firebase
Deployment	Air-Gapped / On-Premise LLM Deployment, Model Serving & Quantization, Docker, Git, CI/CD
Languages & Web	Python, TypeScript, JavaScript, SQL · React, Next.js, React Native, Node.js, FastAPI, Django
Leadership	AI Training & Enablement, Team Building & Mentoring, AI Strategy, Technical Roadmapping

EDUCATION

B.E., Computer Science & Engineering – Chitkara University, Punjab · CGPA 9.3/10 — Sep 2021 – Jul 2025

Certification: Data Science, Sabudh Foundation (2025)